UNIVERSITAS AMIKOM YOGYAKARTA

QS STARS RATING SYSTEM

RANKED 21ST IN THE WORLD WURI The World University Rankings for Innovation in Industrial Application

Strategic Information System Planning

Introduction of Enterprise Architecture



Capaian Pembelajaran MK

- Mampu menjelaskan konsep dan teori dalam perencanaan strategis sistem infromasi sebuah organisasi/enterprise
- Mampu menyusun arsitektur TI secara tepat sesuai dengan proses bisnis yang ada di organisasi/enterprise
- Mampu membuat peta rencana TI hasil yang sesuai dengan visi, misi, dan tujuan organisasi/enterprise

Kondisi Suatu Organisasi

- Ribuan pegawai dan puluhan unit kerja yang mungkin luput dari pengawasan.
- Proses bisnis yang rumit dan berlapis.
- Infrastruktur dan aset yang sulit dikendalikan.
- Staf dan pimpinan baru membutuhkan waktu adaptasi untuk mengenal kondisi organisasi.



Masalah di Organisasi

- Saat satu unit kerja membeli laptop dan printer, ternyata ada unit kerja lain yang memiliki laptop dan printer menganggur.
  - Bagaimana memperkirakan kebutuhan tahunan laptop dan printer?
  - Saat suatu unit kerja mengusulkan pengadaan baru, ternyata barang serupa tersedia secara berlebih di unit kerja lainnya.
- Ketika staf mengajukan cuti:
  - Apa dasar pertimbangan atasan dalam menyetujui permohonan cuti tersebut?
  - Berapa perkiraan jumlah staf yang akan mengajukan cuti tiap bulan?
- Ketika pimpinan memerlukan data trend penjualan:
  - Dokumen apa saja yang diperlukan dan dimana lokasinya?
- Ketika merancang kebijakan keuangan, sering kali tidak akurat karena keputusan tidak berbasis data dan informasi.
  - Bagaimana memperkirakan kebutuhan anggaran tahunan secara akurat?
- Ketika terjadi pergantian pimpinan organisasi:
  - Bagaimana agar pimpinan baru dapat cepat memahami kondisi organisasi?

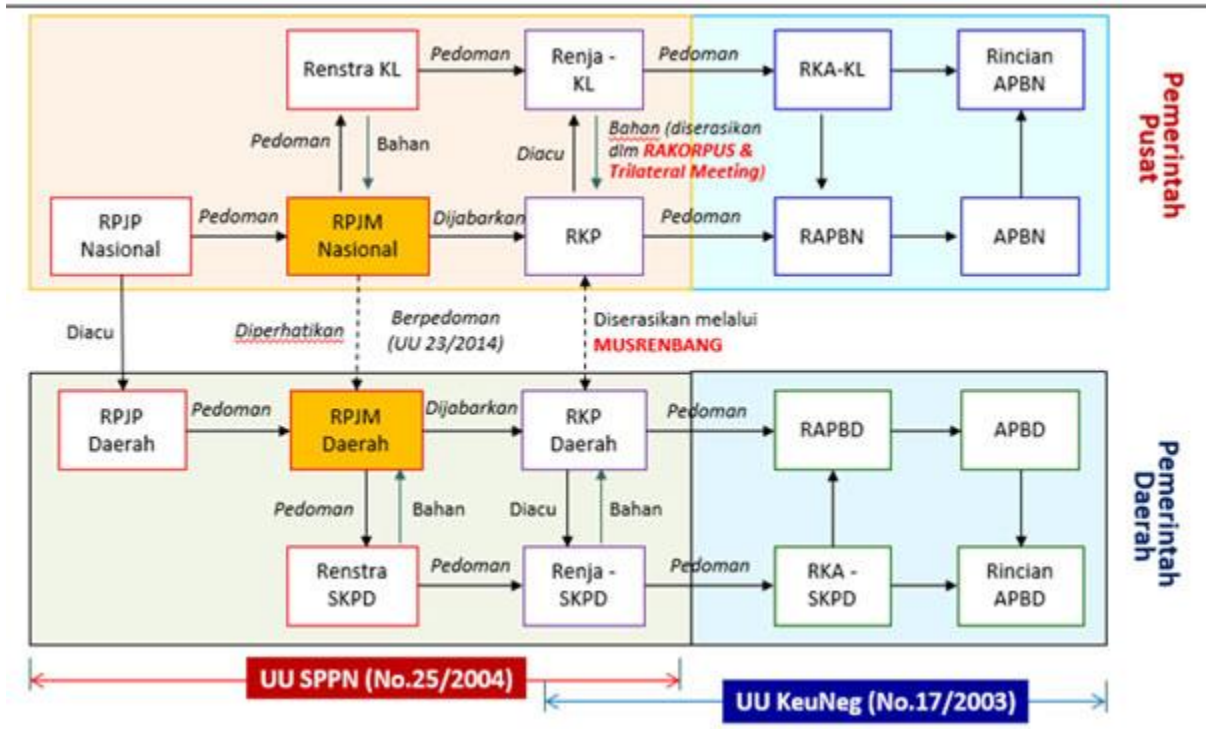
Masalah di Pemerintah

"Apa INDIKATOR RPJMD/Renstra IPPD?"

"APA SASARAN & TUPOKSI OPD/Instansi?"

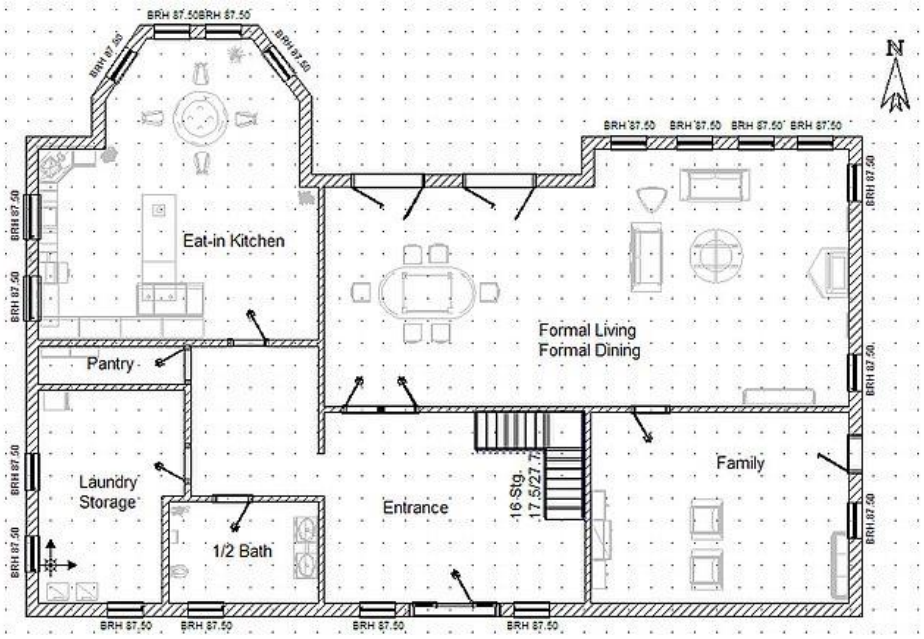
"Apa yang harus DIBANGUN Pemda 5 Tahun Mendatang?"

"Apa yang DIMILIKI Pemda?"



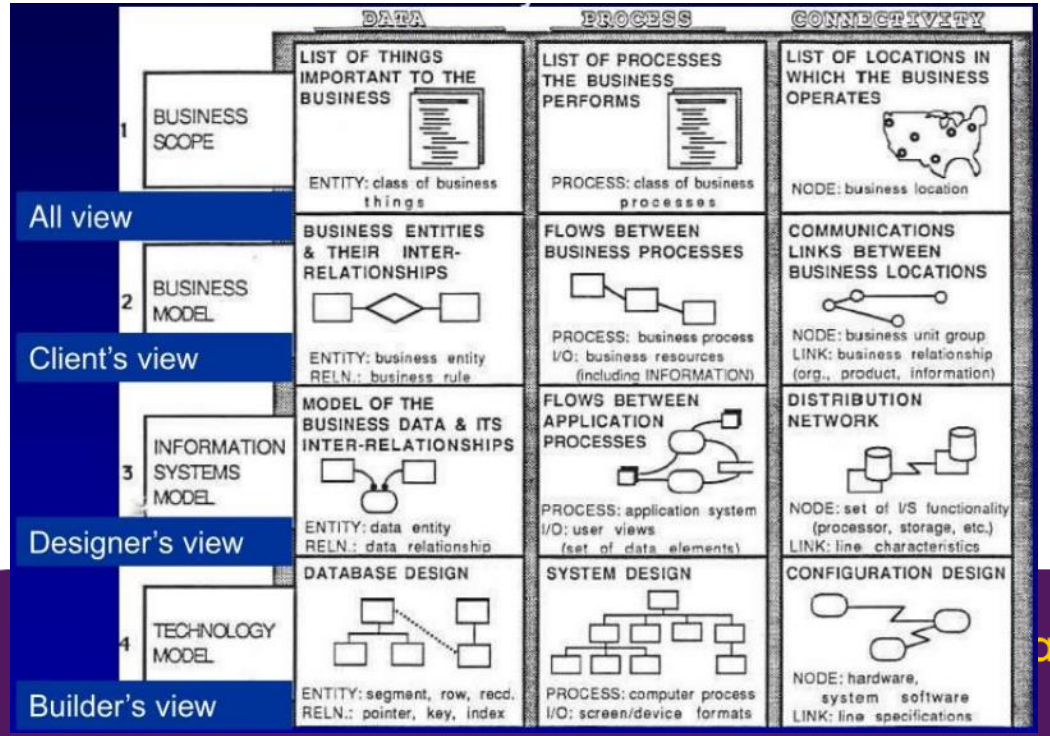
- Struktur Organisasi
- Tujuan Organisasi
- Business Process
- Struktur Data
- Software Sistem
- Infrastruktur IT
- Kebijakan Keamanan
- dsb ...

From this...



1. Business Architecture
2. Data Architecture
3. Application Architecture
4. Technology Architecture

... to this



Background

- Tahun 80-90an, **kompleksitas sistem IT meningkat**, tetapi tidak ada alat untuk mengorganisir artefak arsitektur secara holistik.
- Zachman (IBM) ingin menciptakan "**taxonomy**" yang membantu perusahaan memahami:
  - Apa yang dimiliki (data, proses, aplikasi, dll.).
  - Bagaimana semua komponen berhubungan.
- Ide Kunci:
  - Sistem IT perlu dilihat dari berbagai perspektif (pemilik bisnis, arsitek, developer, dll.).
  - Mengadopsi analogi blueprint arsitektur bangunan untuk TI.
- Hasil: *Framework for Information System Architecture* → Matriks Zachman (6x6) sebagai alat klasifikasi, bukan metodologi

Enterprise is...

- Kumpulan organisasi yang memiliki tujuan bersama, seperti lembaganya pemerintahan, bagian dari sebuah perusahaan, atau keseluruhan perusahaan itu sendiri.
- Perusahaan besar dapat terdiri dari beberapa entitas bisnis.
- Sebuah "perusahaan yang diperluas" dapat mencakup mitra, pemasok, dan pelanggan.

Architecture is...

- The organizational structure of a system or component
  (IEEE Standard 610, Standard Computer Dictionary: A Compilation of IEEE Standard Computer Glossaries)

- The fundamental organization of a system embodied in its components, their relationships to each other, and to the environment, and the principles guiding its design and evolution
  (IEEE Standard 1471, IEEE Recommended Practice for Architectural Description of Software-Intensive Systems)

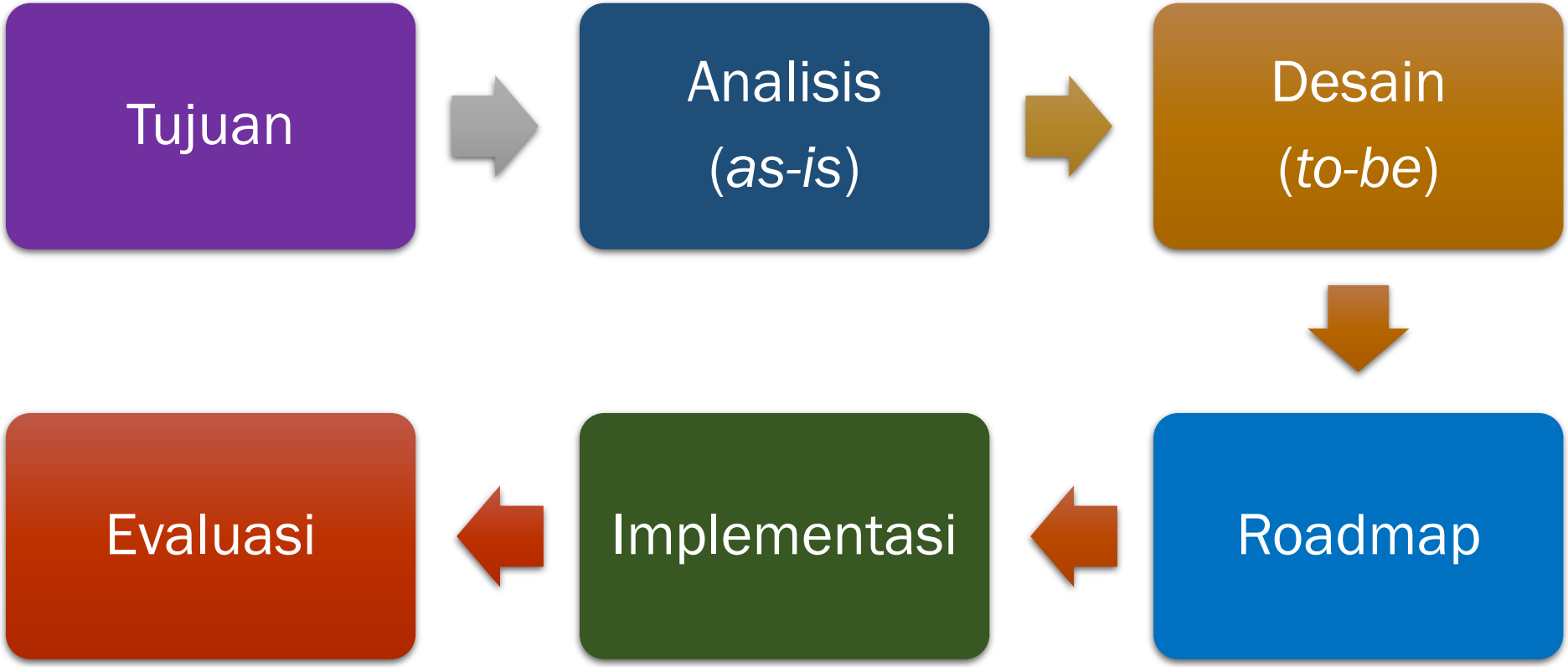
Enterprise Architecture is...

- The organizing logic for business processes and IT infrastructure reflecting the integration and standardization requirements of the firm's operating model

  (MIT Center for Information Systems Research)

- A well-defined practice for conducting enterprise analysis, design, planning, and implementation, using a comprehensive approach at all times, for the successful development and execution of strategy. *(Federation of Enterprise Architecture Professional Organizations)*

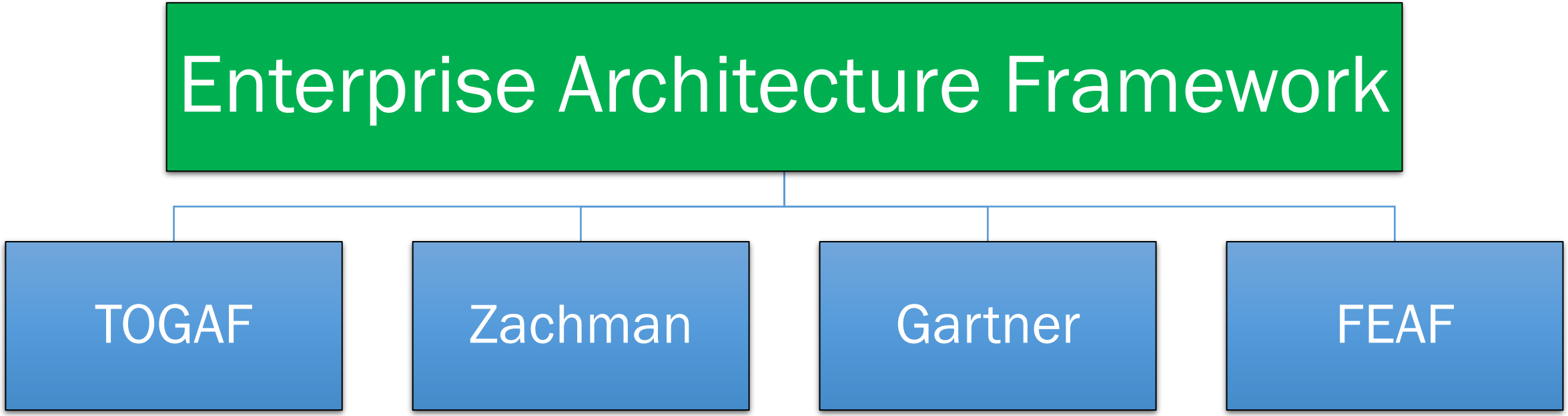
Bagaimana cara membuat EA? *(simple)*



Breakdown

- Definisikan **Tujuan** Bisnis
  - "Integrasikan semua sistem dalam 1 tahun."
- **Analisis** Kondisi Saat Ini (*As-Is*)
  - Buat daftar aplikasi, infrastruktur, dan proses bisnis yang ada.
  - Identifikasi masalah (misal: data terisolasi, duplikasi sistem).
- **Desain** Target Arsitektur (*To-Be*)
  - Gambarkan solusi dengan diagram sederhana (contoh: alur data terintegrasi).
- Buat **Roadmap**
  - Prioritas proyek 1 (Strukturisasi API – 2 bulan)
  - Prioritas proyek 2 (Migrasi server local ke cloud – 2 bulan).... dst
- **Implementasi** & **Evaluasi**
  - Gunakan pendekatan iteratif (agile) untuk penyesuaian berkala.

Bagaimana cara membuat EA? *(hardway)*



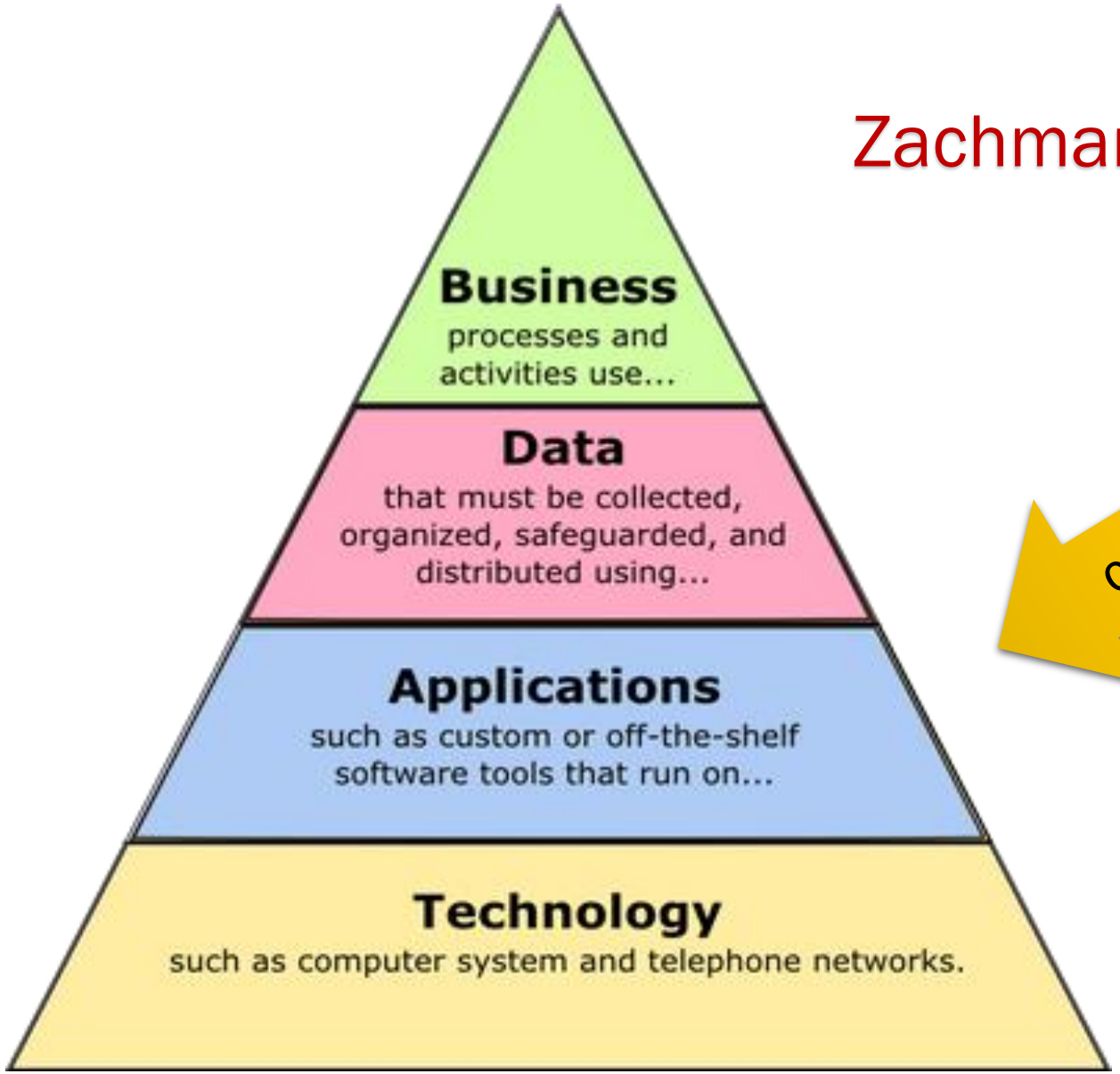
EA Framework is...

- Sebuah **struktur dasar** atau **serangkaian struktur** yang digunakan untuk mengembangkan berbagai arsitektur berbeda:
  - Menggambarkan metode ideal perancangan perusahaan dalam bentuk **blok-blok penyusun**, serta cara blok-blok tersebut saling terhubung.
  - Menyediakan seperangkat alat bantu dan **kosakata** yang seragam.
  - Mencakup daftar **standar rekomendasi dan produk** yang sesuai, untuk mengimplementasikan blok-blok penyusun tersebut.

Mengapa perlu EAF?

- Menggunakan kerangka arsitektur akan mempercepat dan menyederhanakan pengembangan arsitektur,
- Memastikan cakupan solusi yang dirancang lebih menyeluruh,
- Menjamin bahwa arsitektur yang dipilih mampu mengakomodasi pertumbuhan di masa depan sesuai kebutuhan bisnis.

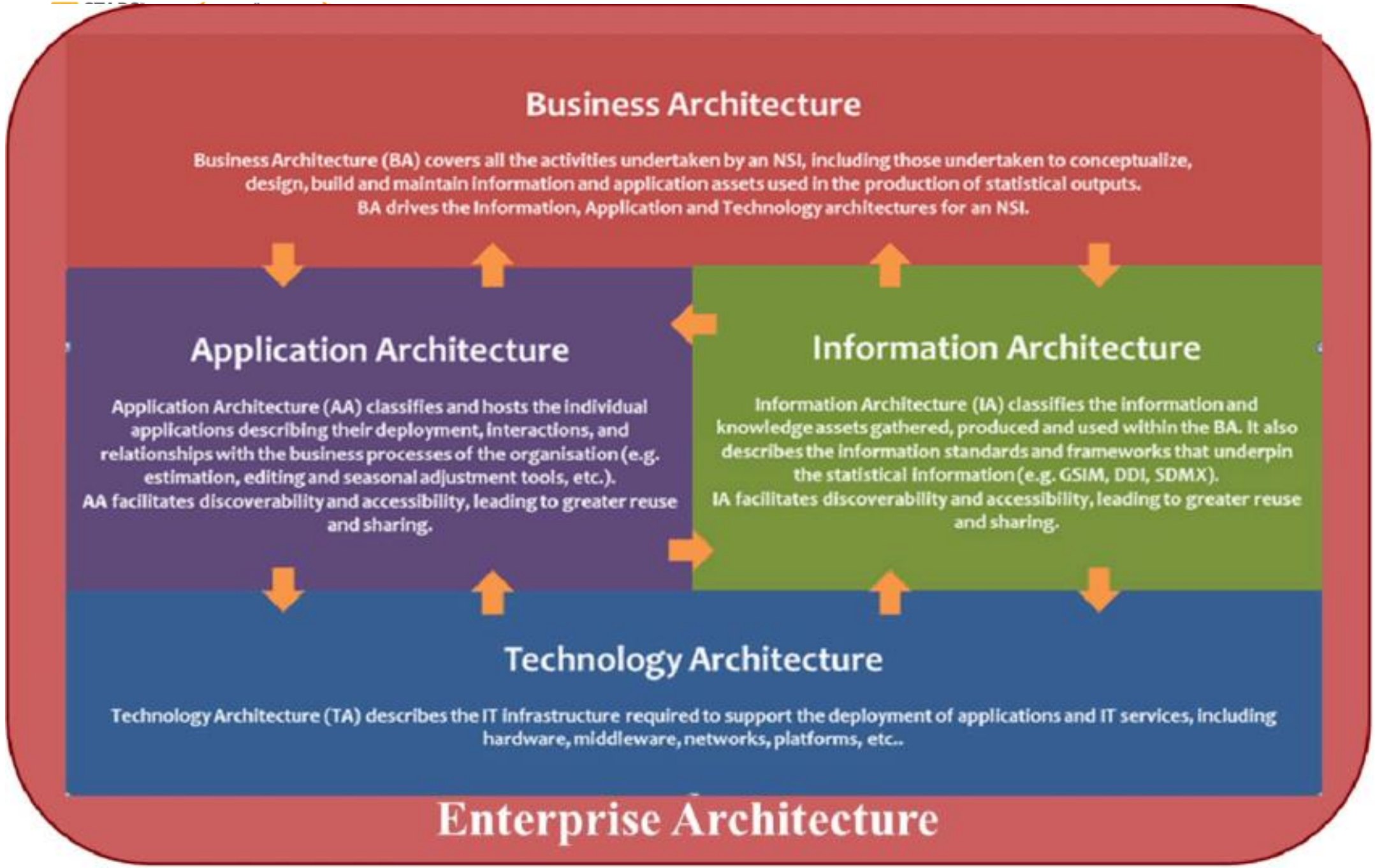
Zachman's Idea



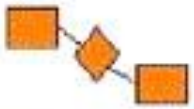
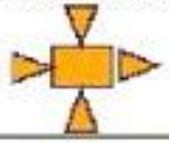
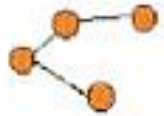
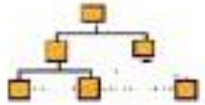
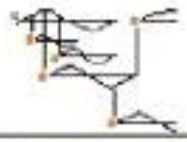
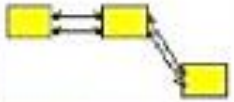
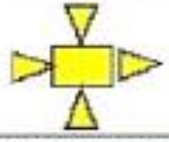
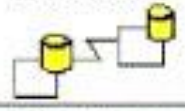
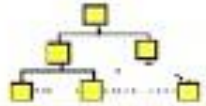
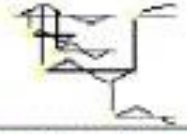
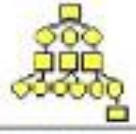
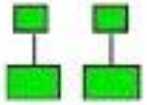
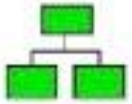
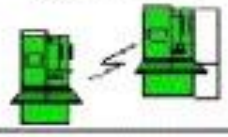
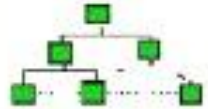
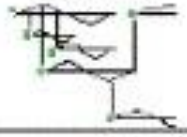
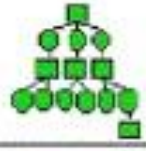
Classification





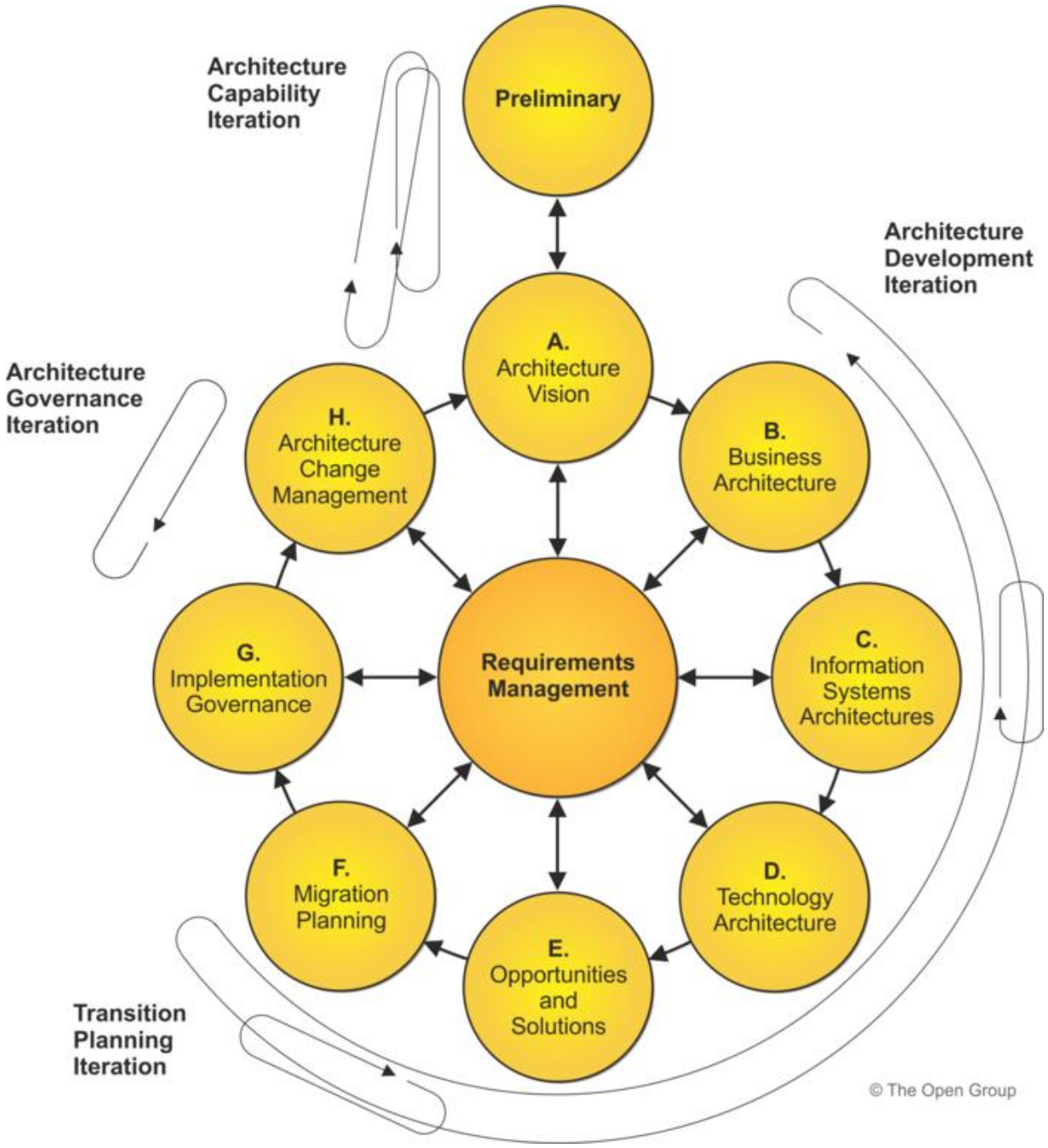

Business Architecture
Business Architecture (BA) covers all the activities undertaken by an NSI, including those undertaken to conceptualize, design, build and maintain information and application assets used in the production of statistical outputs.
BA drives the Information, Application and Technology architectures for an NSI.
Application Architecture
Application Architecture (AA) classifies and hosts the individual applications describing their deployment, interactions, and relationships with the business processes of the organisation (e.g. estimation, editing and seasonal adjustment tools, etc.).
AA facilitates discoverability and accessibility, leading to greater reuse and sharing.
Information Architecture
Information Architecture (IA) classifies the information and knowledge assets gathered, produced and used within the BA. It also describes the information standards and frameworks that underpin the statistical information (e.g. GSIM, DDI, SDMX).
IA facilitates discoverability and accessibility, leading to greater reuse and sharing.
Technology Architecture
Technology Architecture (TA) describes the IT infrastructure required to support the deployment of applications and IT services, including hardware, middleware, networks, platforms, etc..
Enterprise Architecture

Initial ⟷ Extended

	DATA *what*	FUNCTION *How*	NETWORK *Where*	PEOPLE *Who*	TIME *When*	MOTIVATION *Why*	
Objective Scope ***Contextual*** *Role: Planner*	List of Things Important in the Business 	List of core Business Process 	List of Business Locations 	List of Important Organizations 	List of Events 	List of Business Goals/Strategies 	**Objective Scope** ***Contextual*** *Role: Planner*
Enterprise Model ***Conceptual*** *Role: Owner*	Conceptual Data/ Object Model 	Business Process Model 	Business Logistics System 	Work Flow Model 	Master Schedule 	Business Plan 	**Enterprise Model** ***Conceptual*** *Role: Owner*
System Model ***Logica*** *Role: Designer*	Logical Data Model 	System Architecture Model 	Distributed Systems Architecture 	Human Interface Architecture 	Procesing Structure 	Business Role Model 	**System Model** ***Logica*** *Role: Designer*
Technology Model ***Physical*** *Role: Builder*	Physical Data/ Class Model 	Technology Design Model 	Technology Architecture 	Presentation Architecture 	Control Structure 	Rule Design 	**Technology Model** ***Physical*** *Role: Builder*
Detailed Representations ***out of Context*** *Role: Programmer*	Data Definations 	Program 	Network Architecture 	Security Architecture 	Timing Defination 	Rule Specification 	**Detailed Representations** ***out of Context*** *Role: Programmer*
Functioning Enterprise *Role: User*	Usable Data	Working Function	Usable Network	Functioning Organization	Implemented Schedule	Working Strategy	**Functioning Enterprise** *Role: User*

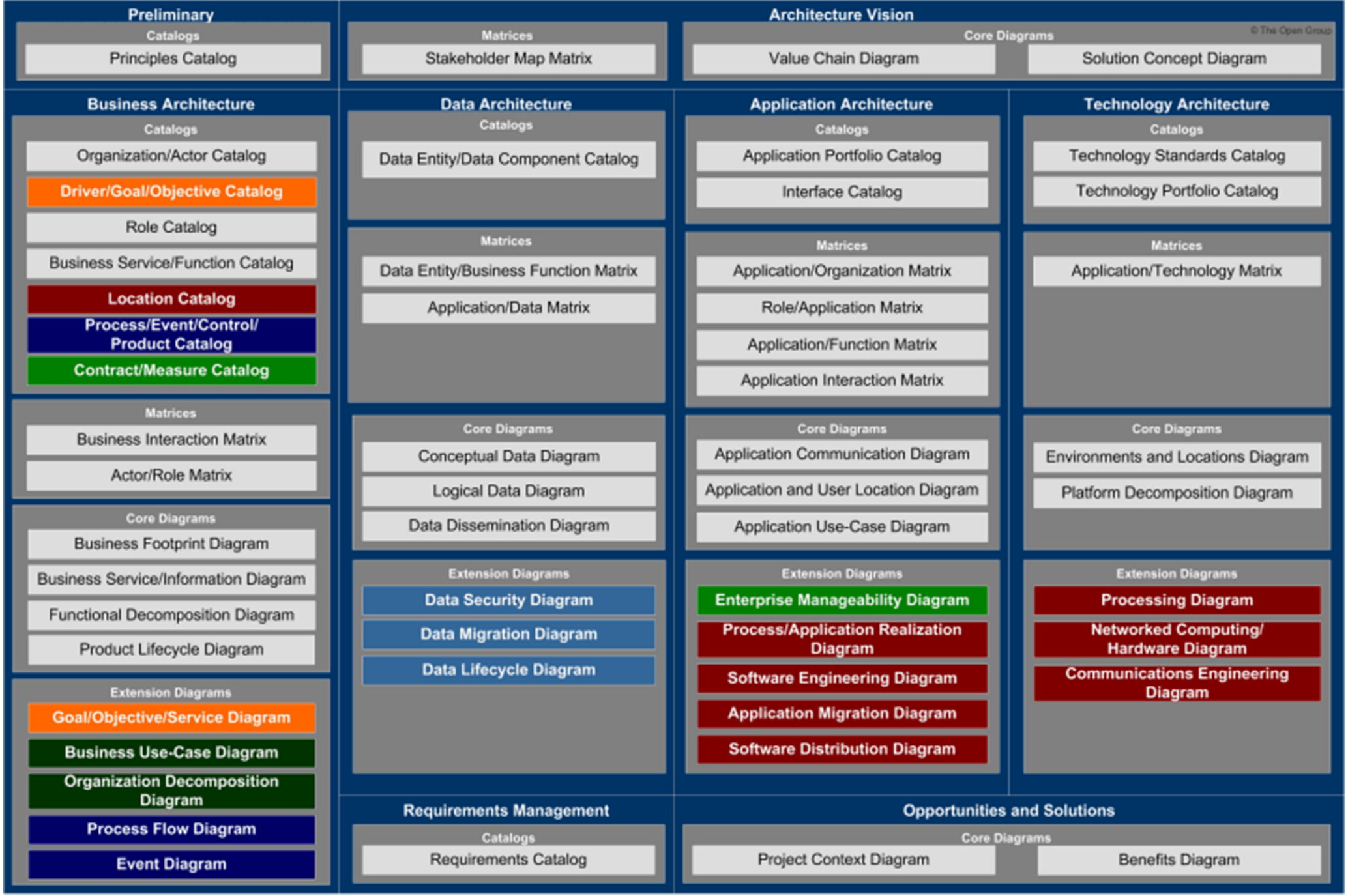
TOGAF – The Open Group Architecture Framework

- Meskipun ZACHMAN cocok untuk pemahaman konseptual, tetapi kurang panduan praktis
- TOGAF lahir dengan standar praktis untuk mengembangkan EA
  - Proyek IT sering gagal akibat kurangnya pendekatan terstruktur.
  - Organisasi butuh panduan reusable untuk align TI dengan bisnis.
- Kunci:
  - ADM (Architecture Development Method) → pengembangan arsitektur secara sistematis
  - Struktur jelas dan terorganisir



Incremental & Iterative !

Architecture Development Method


Preliminary
Catalogs
Principles Catalog
Architecture Vision
Matrices
Stakeholder Map Matrix
Core Diagrams
Value Chain Diagram
Solution Concept Diagram
© The Open Group
Business Architecture
Catalogs
Organization/Actor Catalog
Driver/Goal/Objective Catalog
Role Catalog
Business Service/Function Catalog
Location Catalog
Process/Event/Control/ Product Catalog
Contract/Measure Catalog
Matrices
Business Interaction Matrix
Actor/Role Matrix
Core Diagrams
Business Footprint Diagram
Business Service/Information Diagram
Functional Decomposition Diagram
Product Lifecycle Diagram
Extension Diagrams
Goal/Objective/Service Diagram
Business Use-Case Diagram
Organization Decomposition Diagram
Process Flow Diagram
Event Diagram
Data Architecture
Catalogs
Data Entity/Data Component Catalog
Matrices
Data Entity/Business Function Matrix
Application/Data Matrix
Core Diagrams
Conceptual Data Diagram
Logical Data Diagram
Data Dissemination Diagram
Extension Diagrams
Data Security Diagram
Data Migration Diagram
Data Lifecycle Diagram
Application Architecture
Catalogs
Application Portfolio Catalog
Interface Catalog
Matrices
Application/Organization Matrix
Role/Application Matrix
Application/Function Matrix
Application Interaction Matrix
Core Diagrams
Application Communication Diagram
Application and User Location Diagram
Application Use-Case Diagram
Extension Diagrams
Enterprise Manageability Diagram
Process/Application Realization Diagram
Software Engineering Diagram
Application Migration Diagram
Software Distribution Diagram
Technology Architecture
Catalogs
Technology Standards Catalog
Technology Portfolio Catalog
Matrices
Application/Technology Matrix
Core Diagrams
Environments and Locations Diagram
Platform Decomposition Diagram
Extension Diagrams
Processing Diagram
Networked Computing/ Hardware Diagram
Communications Engineering Diagram
Requirements Management
Catalogs
Requirements Catalog
Opportunities and Solutions
Core Diagrams
Project Context Diagram
Benefits Diagram
Infrastructure Consolidation Extension
Governance Extension
Motivation Extension
Process Modeling Extension
Data Modeling Extension
Services Extension
Core Content

Head-to-head

Aspek	Zachman Framework	TOGAF
Fokus	**Klasifikasi** (Taxonomy) - Memetakan artefak arsitektur ke dalam matriks untuk memberikan gambaran holistik.	**Proses** - Metodologi pengembangan arsitektur (ADM) untuk mencapai target bisnis.
Analog	Seperti "periodic table" untuk EA (statis, deskriptif).	Seperti "cookbook" langkah-langkah membangun EA (dinamis, preskriptif).
Target	Membantu organisasi **mendokumentasikan** dan **memahami** kompo nen EA.	Membantu organisasi **merancang, merencanakan, dan mengimplementasikan** EA.
Use Case	Cocok untuk: - Dokumentasi aset IT. - Align arsitektur dengan bisnis.	Cocok untuk: - Transformasi digital. - Standardisasi TI di perusahaan besar.
Industri	Digunakan di sektor dengan regulasi ketat (misal: pemerintah, kesehatan).	Populer di korporasi (banking, telekomunikasi, retail).
Output	Matriks hubungan entitas (misal: proses bisnis vs. aplikasi).	Dokumen arsitektur, roadmap, dan rencana migrasi.

Head-to-head

Aspek	Zachman Framework	TOGAF
Kurva Belajar	Mudah dipahami secara konseptual (hanya matriks), tetapi sulit diterapkan tanpa panduan.	Lebih kompleks (karena proses ADM), tetapi lebih terstruktur untuk eksekusi.
Tool Support	Kurang tool khusus; sering dipadukan dengan pemodelan (ArchiMate).	Banyak tool pendukung (misal: Sparx EA, BizzDesign).
Fleksibilitas	Sangat fleksibel (hanya kerangka klasifikasi).	Terstruktur namun bisa dimodifikasi (ADM bisa disesuaikan).
Format	Matriks 6x6: - **Baris**: Perspektif (Scope, Business, System, dll.). - **Kolom**: Pertanyaan (What, How, Where, Who, When, Why).	**ADM (Architecture Development Method)**: 10 fase iteratif (Visi, Bisnis, Aplikasi, Teknologi, dll.).
Artefak	Fokus pada **what exists** (as-is) dan **what should be** (to-be).	Fokus pada **how to get there** (to-be, roadmap, gap analysis).
Kelengkapan	Tidak menyediakan panduan implementasi.	Menyediakan template, toolkit, dan best practices.

EAF Ratings

	Ratings			
Criteria	Zachman	TOGAF	FEA	Gartner
Taxonomy completeness	4	2	2	1
Process completeness	1	4	2	3
Reference-model guidance	1	3	4	1
Practice guidance	1	2	2	4
Maturity model	1	1	3	2
Business focus	1	2	1	4
Governance guidance	1	2	3	3
Partitioning guidance	1	2	4	3
Prescriptive catalog	1	2	4	2
Vendor neutrality	2	4	3	1
Information availability	2	4	2	1
Time to value	1	3	1	4

(Roger Sessions, A Comparison of the Top Four EA Methodologies, 2007)

EA Tools

Tier 1: Profesional (Enterprise Grade)

Sparx Systems Enterprise Architect
https://sparxsystems.com
- Pemodelan lengkap (ArchiMate, BPMN, UML), simulasi, code generation.

BizzDesign HoriZZon
https://www.bizzdesign.com
- Collaborative EA, integrasi dengan ADO, Jira, dan Power BI.

Ardoq https://www.ardoq.com
- EA berbasis data, analytics, dan AI-driven insights.

Tier 2: Menengah (Balance Fitur & Harga)

Visual Paradigm
https://www.visual-paradigm.com
- Support ArchiMate, TOGAF ADM, dan Agile.

Software AG Alfabet
https://www.softwareag.com
- PlanningIT untuk transformasi digital.

Avolution ABACUS
https://www.avolution.com
- Analytics berat untuk optimasi biaya TI.

Tier 3: Gratis / Open Source

Archi https://www.archimatetool.com
- Pemodelan ArchiMate dasar, ringan.

Modelio https://www.modelio.org
- UML + BPMN + TOGAF support.

Draw.io
https://app.diagrams.net
- Diagram EA sederhana (tanpa repository).

Rekomendasi Prioritas

- ✓Perusahaan Besar ⟶ Sparx EA atau BizzDesign
- ✓UMKM/Startup ⟶ Visual Paradigm atau Archi
- ✓Cuma Butuh Diagram ⟶ Draw.io
- ✓Butuh simulasi ⟶ Sparx EA atau BizzDesign
- ✓Butuh kolaborasi tim ⟶ LeanIX atau Ardoq

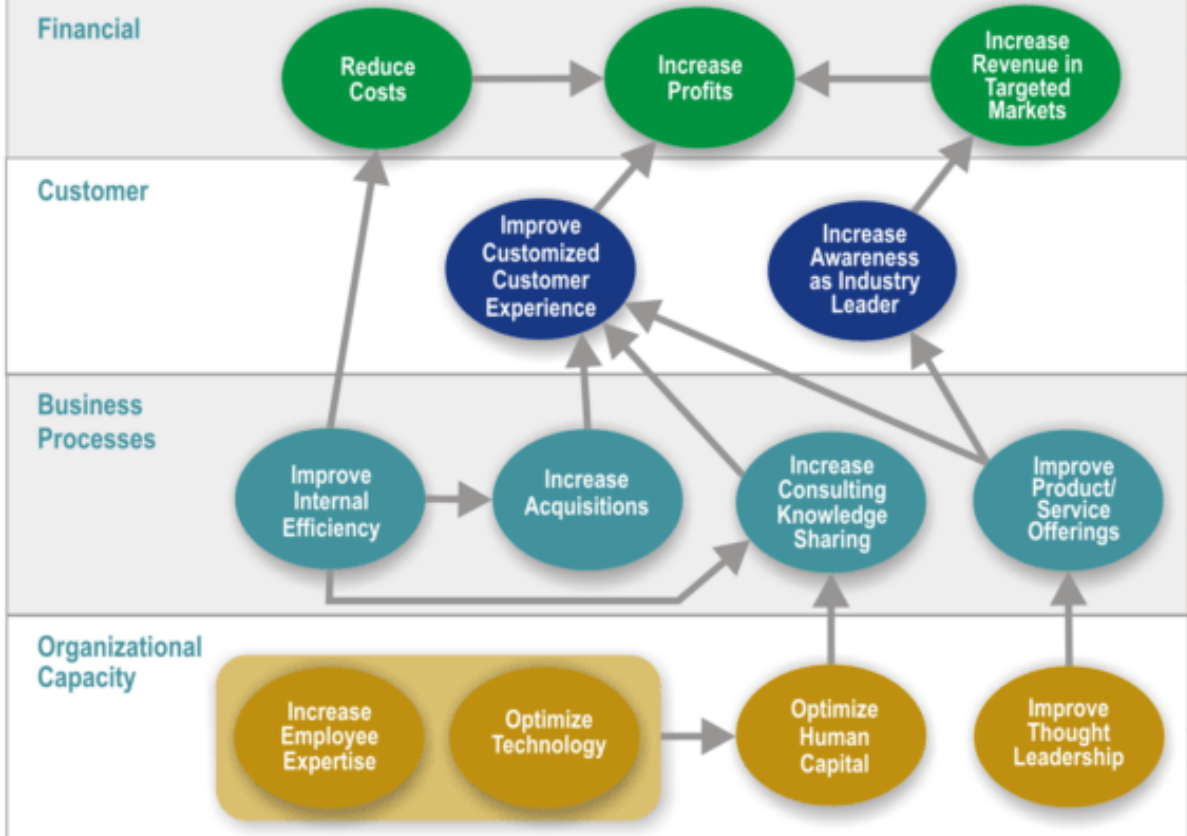
Key Competencies to Create an Enterprise Architecture

1. Enterprise Architecture Framework
2. Enterprise Architecture Tools
3. Business Strategy and Organization Analysis → Business Model Canvas
4. Business Process Model and Notation (BPMN)
5. Data Modeling
6. Unified Modeling Language (UML)
7. Infrastructure and Network Design

BUSINESS MODEL CANVAS



Vision:	Be a world-revolutionizing provider of energy products		
Mission:	Efficiently deliver the highest standard of service providing energy products and expertise to our customers		
Strategic Themes:	*International Growth*	*Operational Excellence*	*Energy Leadership*
Strategic Results:	Increase in size and shareholder value through acquisitions, organic growth and asset optimization.	Process and services are executed in a timely, accurate, value-added and cost-effective manner that exceeds stakeholder expectations.	Worldwide recognition for thought leadership and energy services expertise.

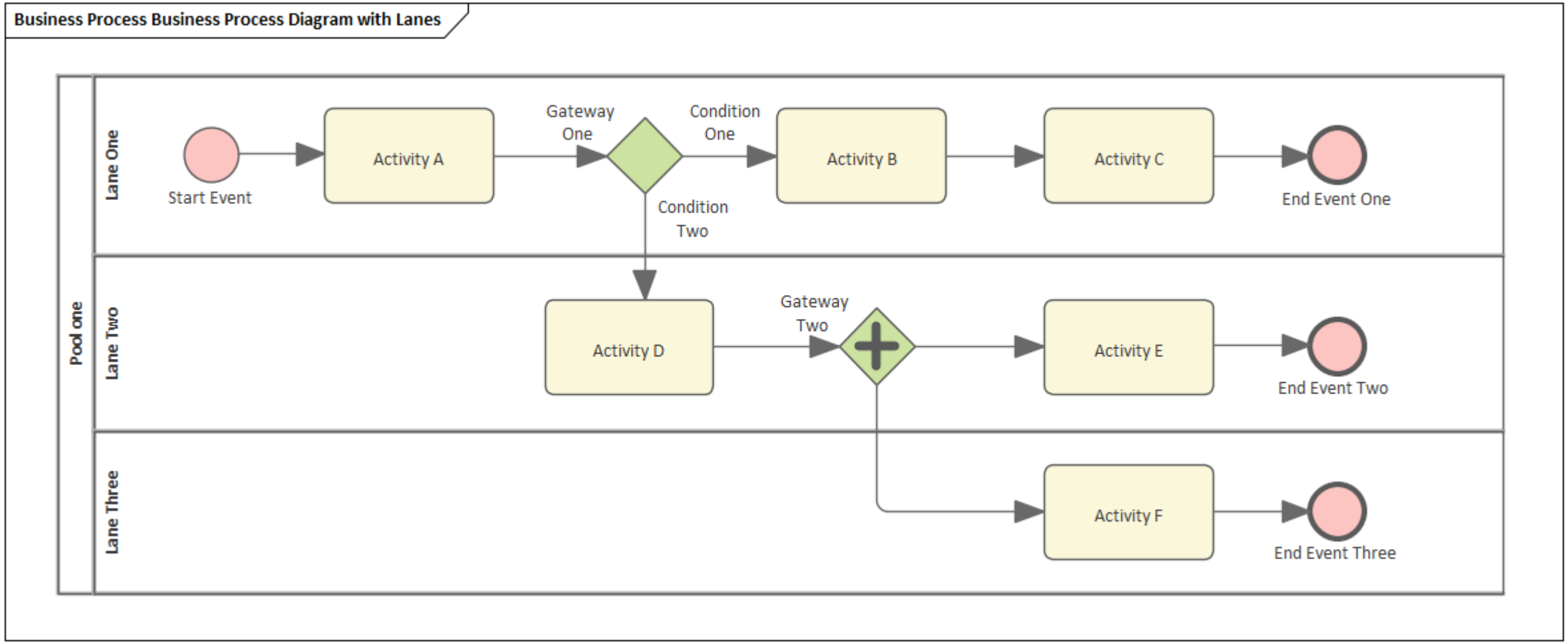


Strategic Objectives and Strategy Map	Measures	Targets	Initiatives
Financial	• Profit • Operating costs • Revenue in targeted markets	• ↑15 % per year • ↓7 % per year • ↑5 % per year	• Implement new financial system and benchmark results
Customer	• Customer experience survey score • % of customers with completed CK charter • Awareness and leadership score • References in media	• ↑5 % improvement in next period • 87 % next period • 98 % next period • 15 new references	• Marketing campaign in new target markets • Develop customer knowledge (CK) charter
Business Processes	• Efficiency index • EBIT from acquisitions • Time to post • % of transactions in CRM • New products/services as % of total sales	• 87% next period • $15M this year • 5 business days • 65% next period • 14% this year	• Establish acquisition integration team • Establish Business Process Reengineering (BPR) team • Formalize new product development cycle • Formalize Customer Relationship Management (CRM) qualification process
Organizational Capacity	• % employee development plans in place • % systems automated • Operating cost/FTE • Articles published	• 70% this year; 100% by year 6 • 72% this year • $3,350/person this year • 125 this year	• CRM system training • Redesign employee certification process • Establish thought leadership committee • Formalize research expertise strategy

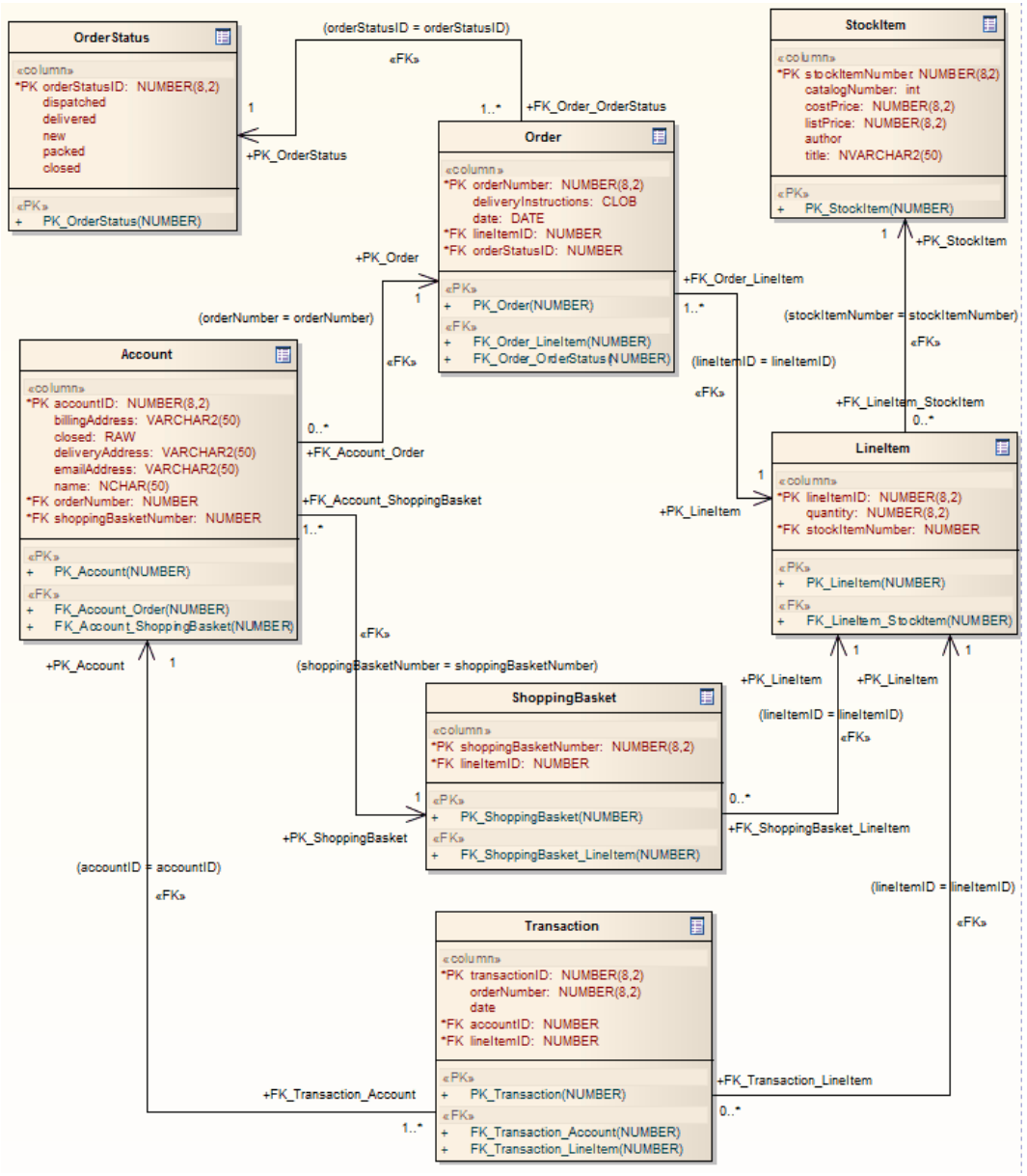
• Integrity • Commitment to Excellence • Customer Focused • Diversity of Knowledge • Honesty • Teamwork/Collaboration • Entrepreneurial • Thirst for Knowledge

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Business Process Model and Notation (BPMN)

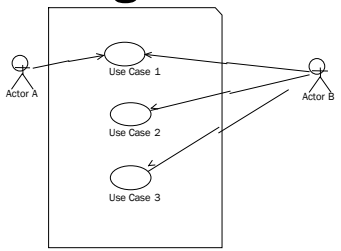


Data Modeling

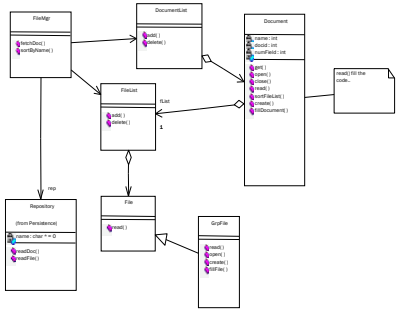


UML

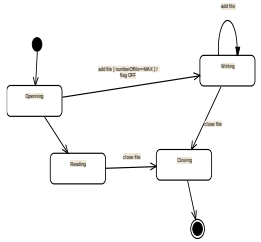
Use-Case Diagram



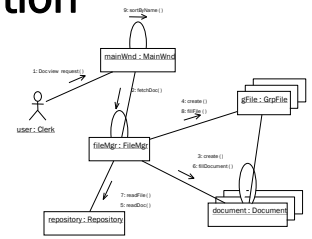
Class Diagram



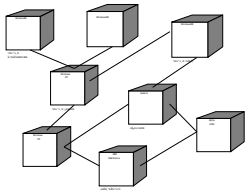
Statechart Diagram

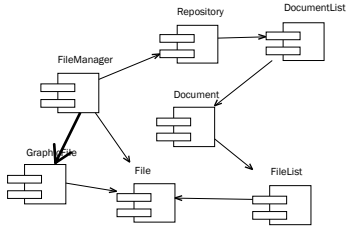


Collaboration Diagram

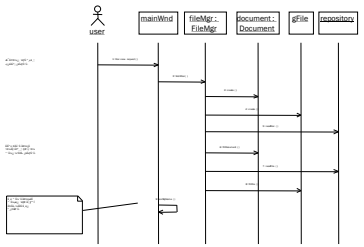


Deployment Diagram



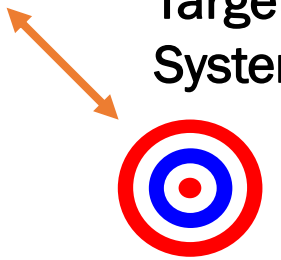


Component Diagram



Sequence Diagram

Forward and Reverse Engineering



Target System